



Predicting HIV-1 Nucleoside Reverse Transcriptase Inhibitor (NRTI) Resistance using RT Mutation Patterns and Interpretable Machine Learning

ResiMap Research Group

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HIV EPIDEMIC ESTIMATES – KENYA 2026

Key HIV indicators and progress compared to 2025



2026 Estimates

2025 Estimates

1. PEOPLE LIVING WITH HIV (PLHIV)



1,481,853
(all ages)

1,326,336



Increase compared to 2025

2. HIV PREVALENCE (15–49 YEARS)



3.22%

3.03%



Increase compared to 2025

3. NEW HIV INFECTIONS



13,936

19,991



Decrease compared to 2025

4. AIDS-RELATED DEATHS



19,434

21,007



Decrease compared to 2025

5. PMTCT COVERAGE



91.1%

90.87%



Increase compared to 2025

6. MTCT RATE



8.0%

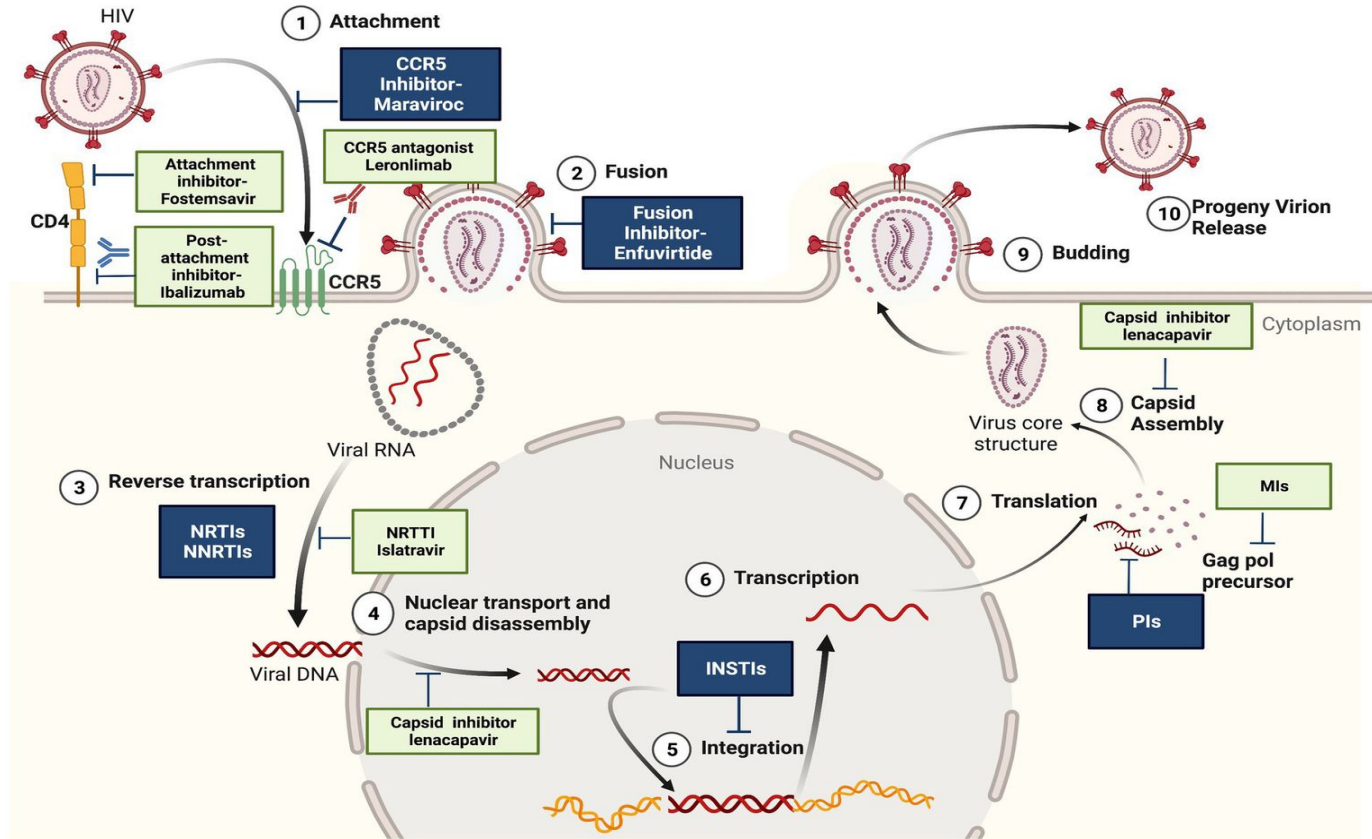
9.26%



Decrease compared to 2025



Targets for HIV-1 Treatment





First-Line ART Regimens in Kenya

Paediatric and Low-Weight Regimens

Birth to 4 Weeks



Use AZT + 3TC + NVP with weight-based dosing for all weights.



Weight-based dosing

4 Weeks to <15 Years (<30 kg)



Use ABC + 3TC + DTG with weight-based dosing for children under 30kg.



Weight-based dosing

Adult and Adolescent Standard Regimens

Standardised TDF + 3TC + DTG Regimen



Prescribed for all patients ≥ 15 years or those < 15 years weighing ≥ 30 kg.



Fixed-Dose Combination



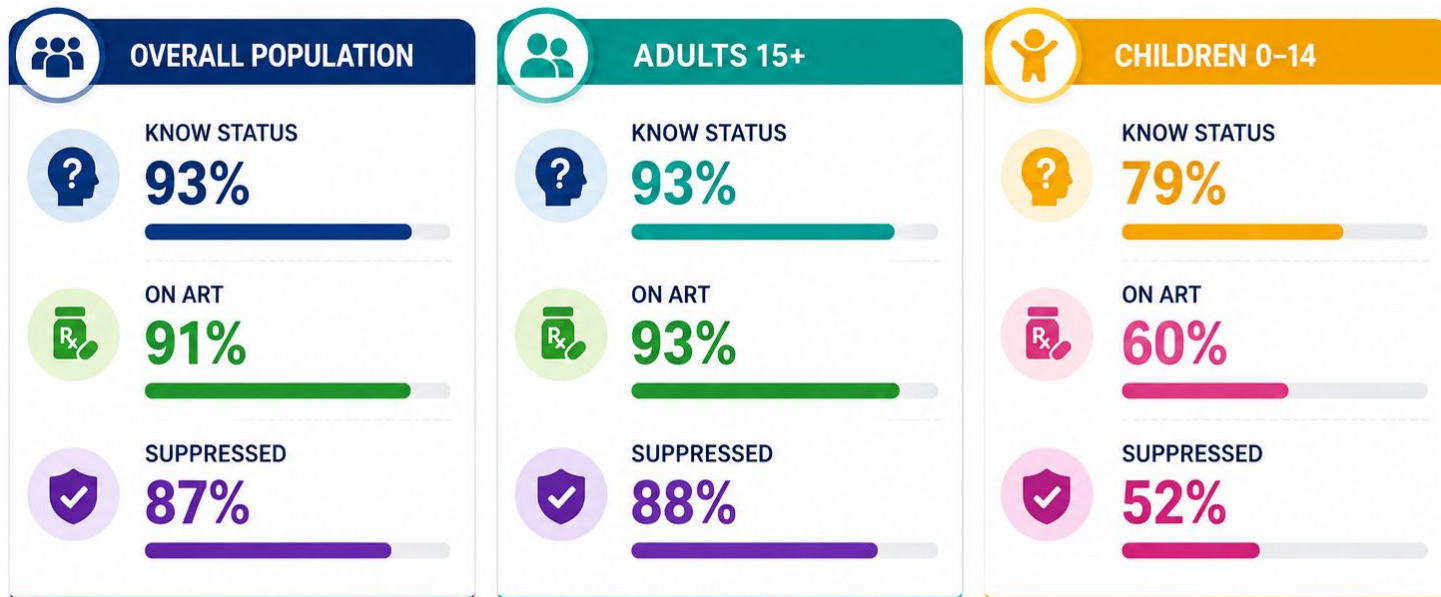
One daily tablet containing TDF (300mg), 3TC (300mg), and DTG (50mg).

1 Tablet Daily

Patient Group	Preferred Regimen	Dosage
Birth to 4 Weeks	AZT + 3TC + NVP	Weight-based
4 Weeks to <15 Years (<30kg)	ABC + 3TC + DTG	Weight-based
Adults & Adolescents (≥ 30 kg)	TDF + 3TC + DTG	1 Tablet Daily



Kenya: National Treatment Cascade



Kenya is making strong progress towards the 95-95-95 targets.

95%

of people living with HIV
know their status

95%

of those diagnosed
are on treatment

95%

of those on treatment
are virally suppressed

Questions

- Which HIV-1 RT mutations are most associated with NRTI resistance?
- Can ML accurately predict NRTI resistance from mutation profiles?
- Which ML algorithm provides the best balance between predictive accuracy and interpretability?
- Do the mutations identified by ML correspond to known biological mechanisms of NRTI resistance?

Specific Aims

- To identify RT mutations associated with NRTI resistance using the SDRDB
- To develop ML model to predict NRTI resistance from HIV RT mutation profiles
- To compare predictive performance and interpretability of ML algorithms for resistance prediction
- To discuss implications for regimen selection in Kenya.

Approach

- Data Extraction
- Model Training
- Algorithm Evaluation
- Clinical Translation

Aim 1: Identifying RT mutations associated with NRTI resistance



- Data Source: Stanford DB
- Search Criteria:

Query interphase: Detailed Phenotype QUery

Virus : HIV1

Gene: Reverse Transcriptase

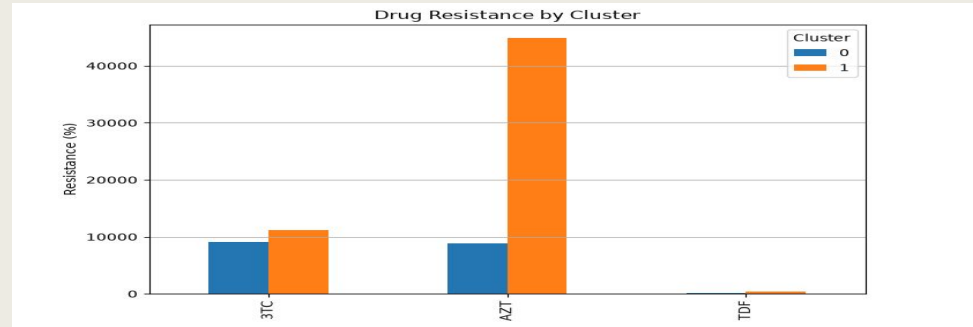
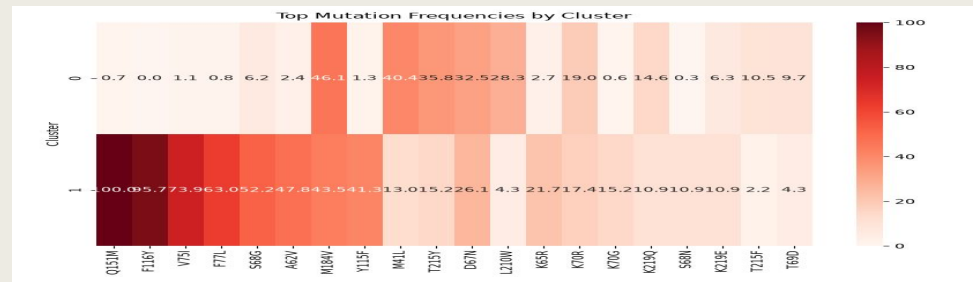
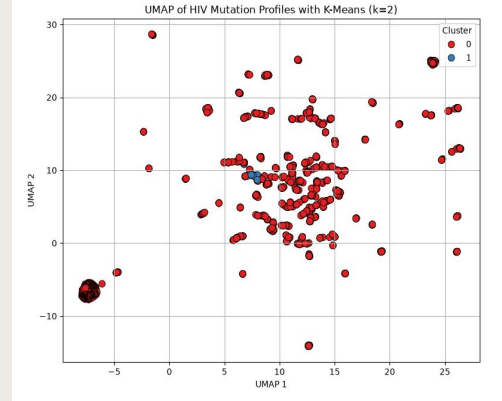
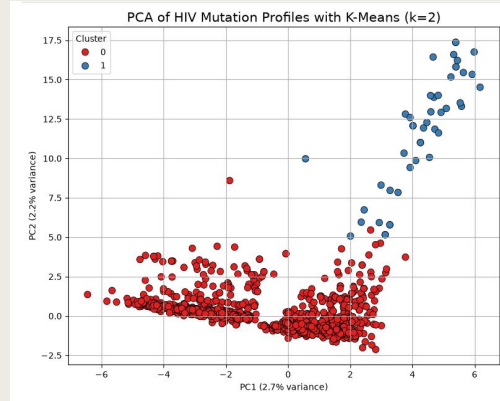
Codons: No Restriction

Susceptibility Test Method: Phenosense

Drugs: All NRTIs

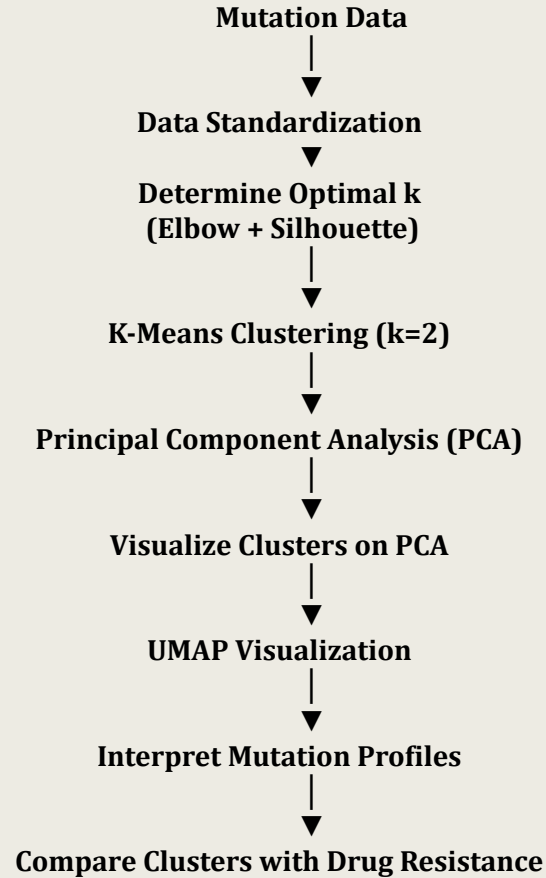
- Drugs studied : Lamivudine(3TC), Tenofovir (TDF), Abacavir (ABC), Zidovudine (AZT)

Exploratory Data Analysis: PCA, K-Means and UMAP





Complete EDA Workflow

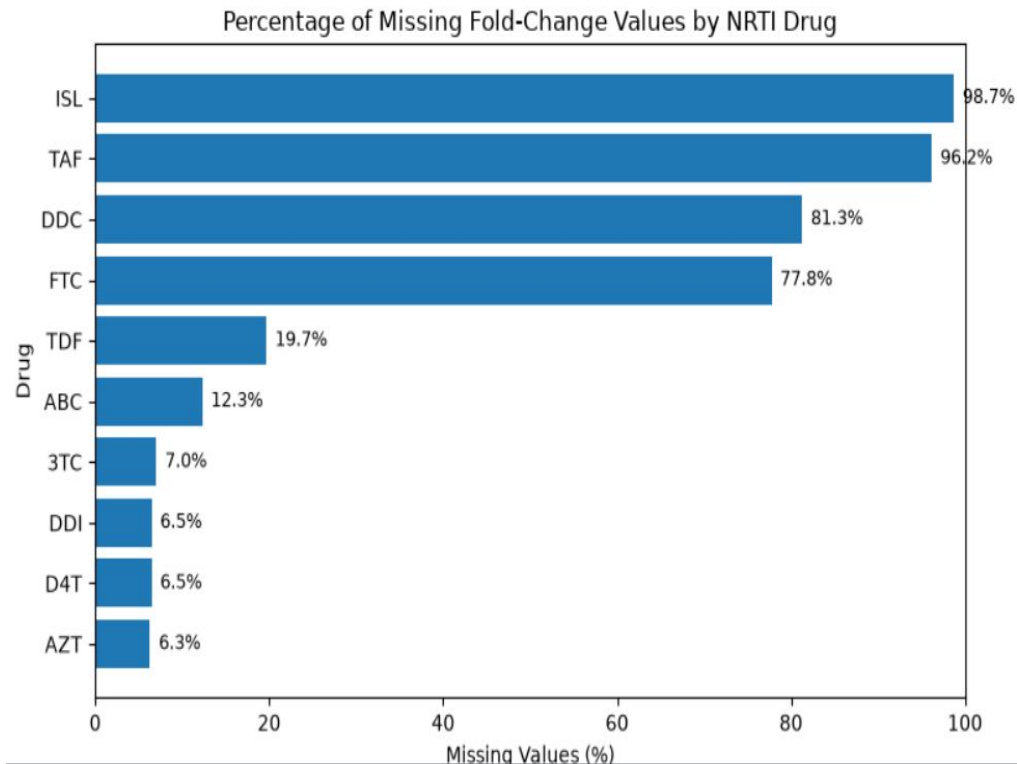


Data Cleaning, Exploration & Standardisation



Data Dimensionality Before : (2505, 335), After:(2505,23)

DRUG	Isolates	Mutations
3TC	2359	42
TDF	2012	42
ABC	2231	42
AZT	2381	42



Optimal Cluster Identification

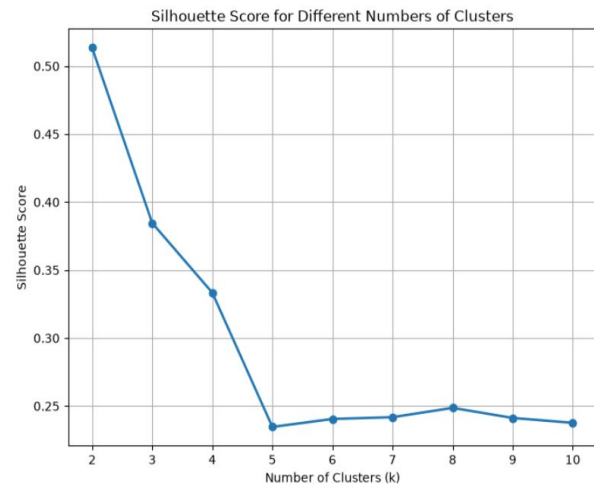
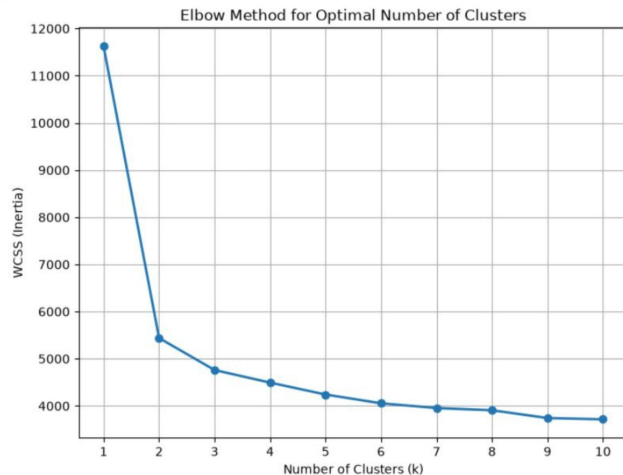
Elbow Method

The within-cluster sum of squares (inertia) curve shows a clear inflection point at $k = 2$, indicating that adding further clusters yields diminishing returns in explanatory power.

Silhouette Score Validation

A silhouette score of **0.51** at $k = 2$ confirms moderate-to-strong cohesion within clusters and separation between them — the highest score across all tested values of k .

Both independent metrics converge on $k = 2$ as the statistically optimal solution.



Cluster Validation

K-Means Partitioning

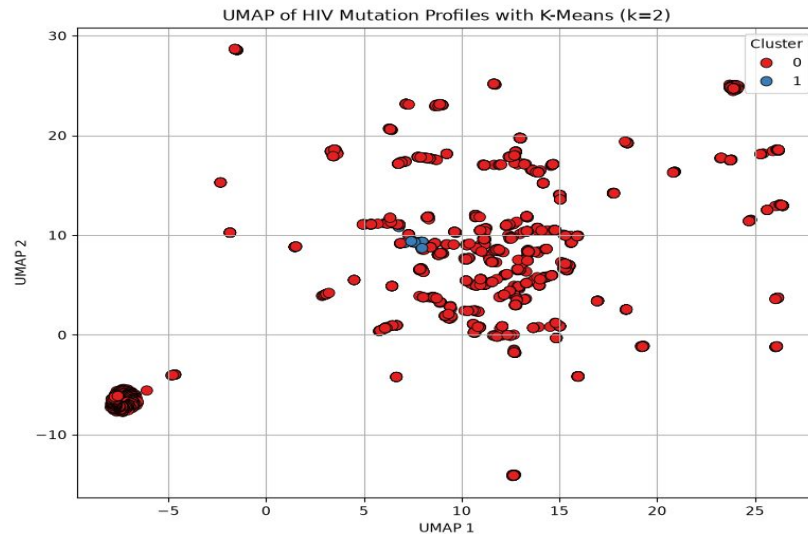
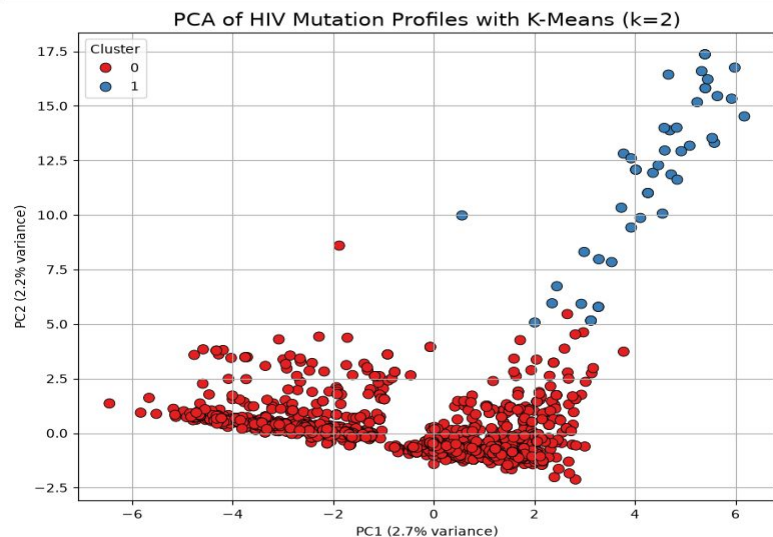
K-Means clustering successfully partitioned HIV isolates into two distinct, non-overlapping groups based on mutation feature profiles.

Principal Component Analysis (PCA)

(PCA reveals clear spatial separation between clusters along PC1 and PC2, confirming structural distinctness in reduced dimensions.

UMAP Confirmation

Dimensionality reduction independently confirms the two-cluster topology, reinforcing that the separation reflects genuine biological structure.





Top Mutations in Each Cluster

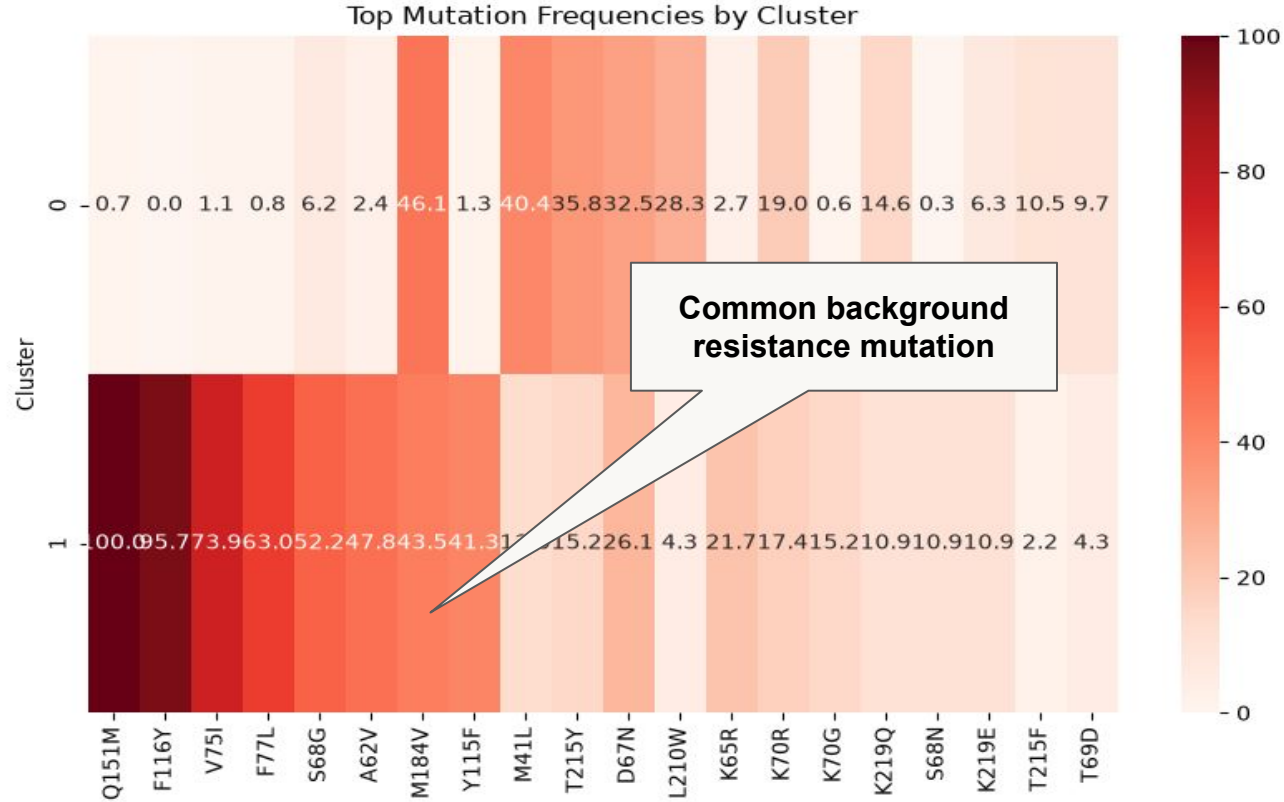
Rank	Top Mutations in Cluster 0	Frequency (%)	Top Mutations in Cluster 1	Frequency (%)
1	M184V	46.07	Q151M	100.00
2	M41L	40.38	F116Y	95.65
3	T215Y	35.78	V75I	73.91
4	D67N	32.49	F77L	63.04
5	L210W	28.26	S68G	52.17
6	K70R	18.95	A62V	47.82
7	K219Q	14.59	M184V	43.47
8	T215F	10.49	Y115F	41.30
9	T69D	9.71	D67N	26.08
10	E44D	8.58	K65R	21.73
11	L74V	7.76	K70R	17.39
12	K219E	6.26	K70G	15.21
13	S68G	6.18	T215Y	15.21

Drug Resistance Pathways

Drug Resistance pathway	Cluster 0	Cluster 1
TAM*-1	★★★★★ Dominant	★ Low
TAM-2	★★★★★ Dominant	★★ Moderate
M184V pathway	★★★★ High	★★★★ High
K65R pathway	★ Very low	★★★★ High
Q151M complex	Not detected	★★★★★ Dominant
Codon 69 insertion	Not detected	Not detected

* Thymidine Analog Mutations

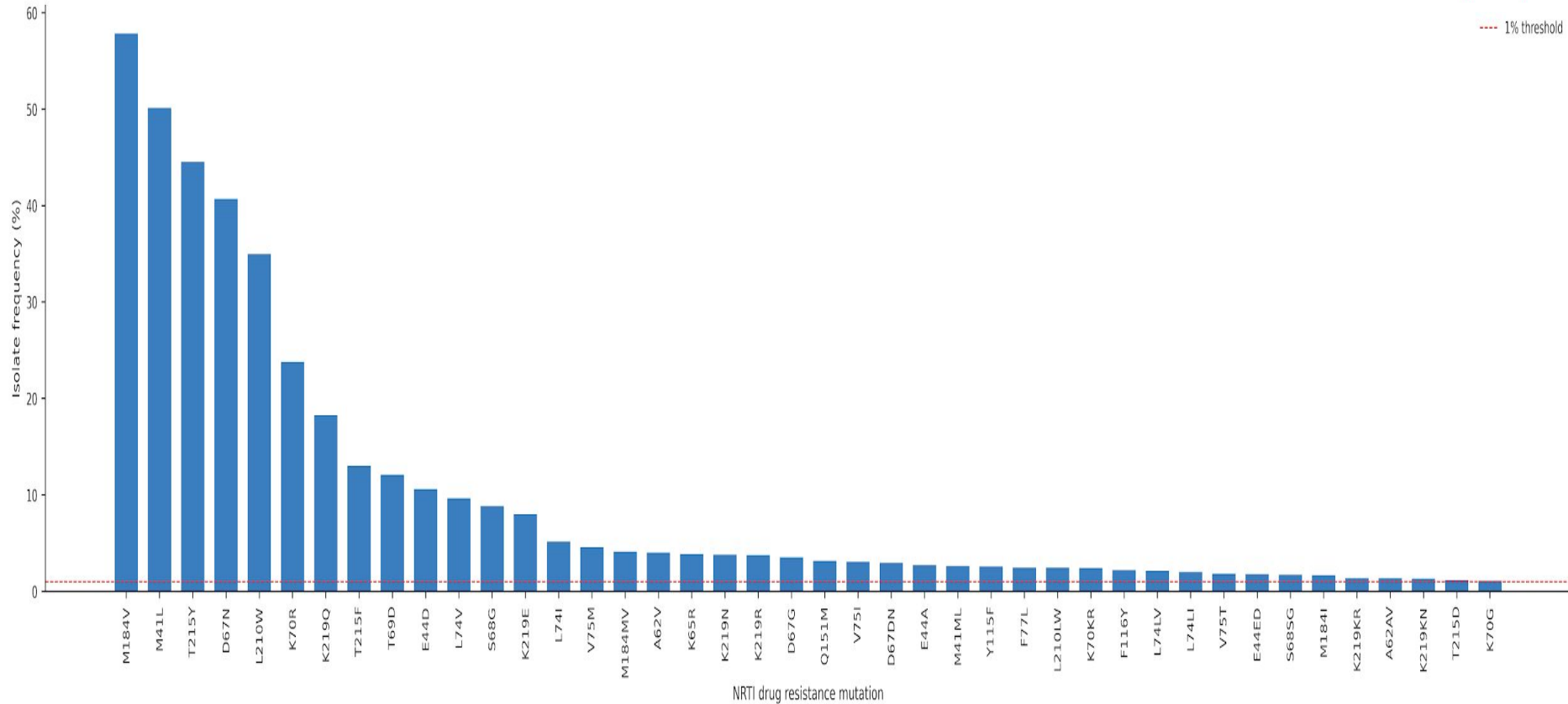
Top Mutation Frequencies by Cluster Heatmap



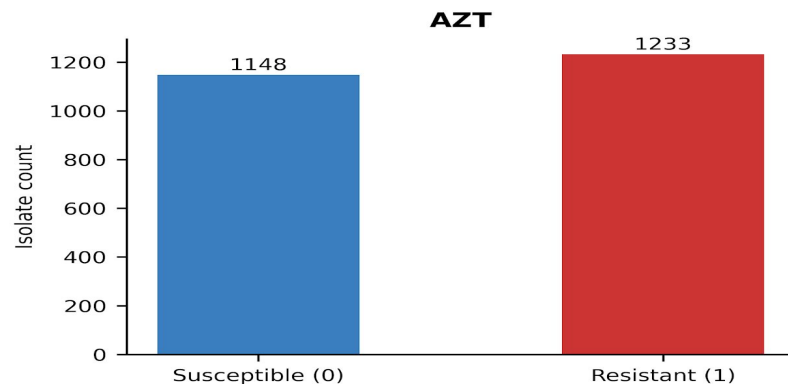
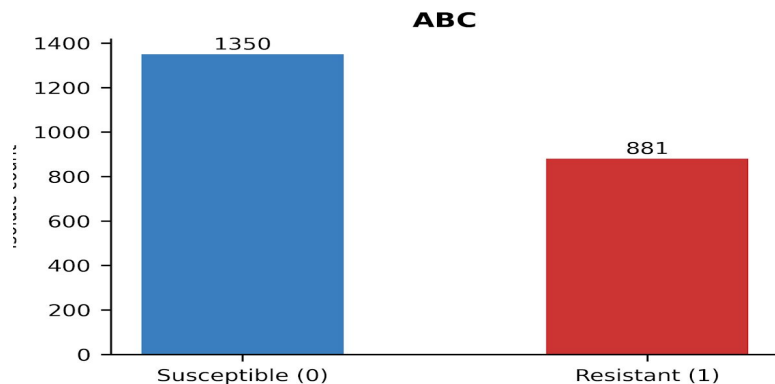
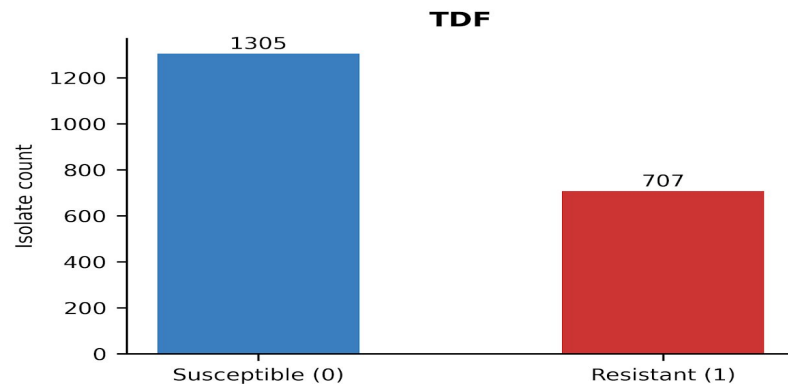
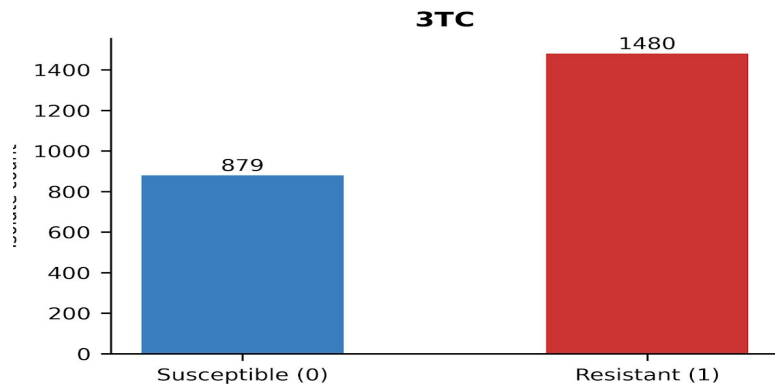


Aim 2: Model Training

Mutation (Feature) Selection: Frequency



Resistance Distribution (**class imbalance**)



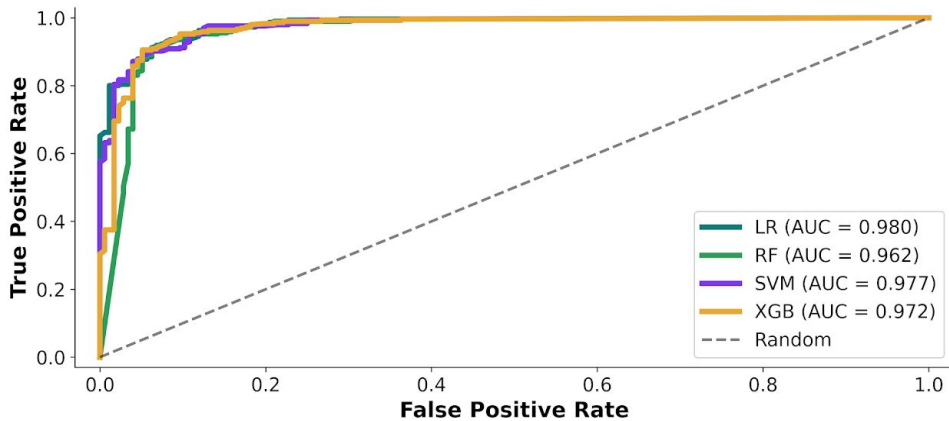


Aim 3: Predictive Performance and Interpretability

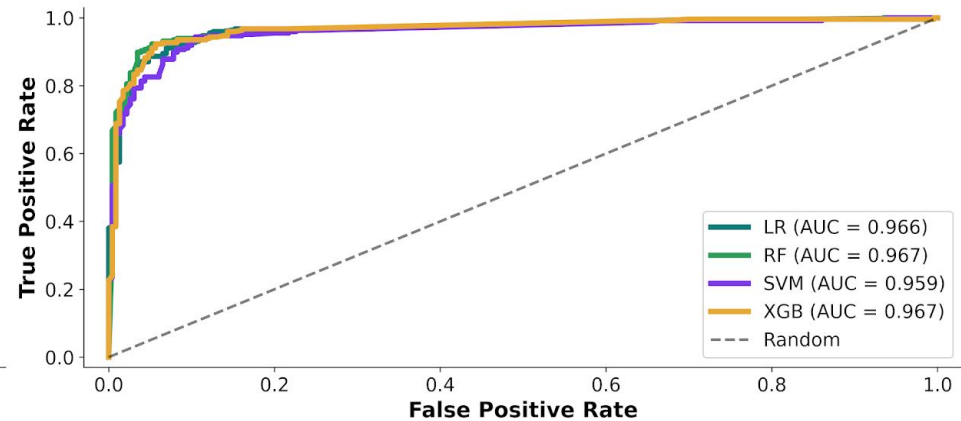
Drug	Model	Precision	Recall	F1_score
3TC	LR	0.91	0.92	0.91
3TC	RF	0.91	0.92	0.92
3TC	SVM	0.9	0.91	0.91
3TC	XGB	0.93	0.92	0.93
ABC	LR	0.87	0.87	0.87
ABC	RF	0.86	0.87	0.87
ABC	SVM	0.87	0.87	0.87
ABC	XGB	0.88	0.88	0.88
TDF	LR	0.83	0.85	0.83
TDF	RF	0.78	0.79	0.78
TDF	SVM	0.8	0.83	0.81
TDF	XGB	0.8	0.81	0.8
AZT	LR	0.92	0.92	0.92
AZT	RF	0.93	0.94	0.93
AZT	SVM	0.91	0.91	0.91
AZT	XGB	0.93	0.93	0.93

ROC-AUC

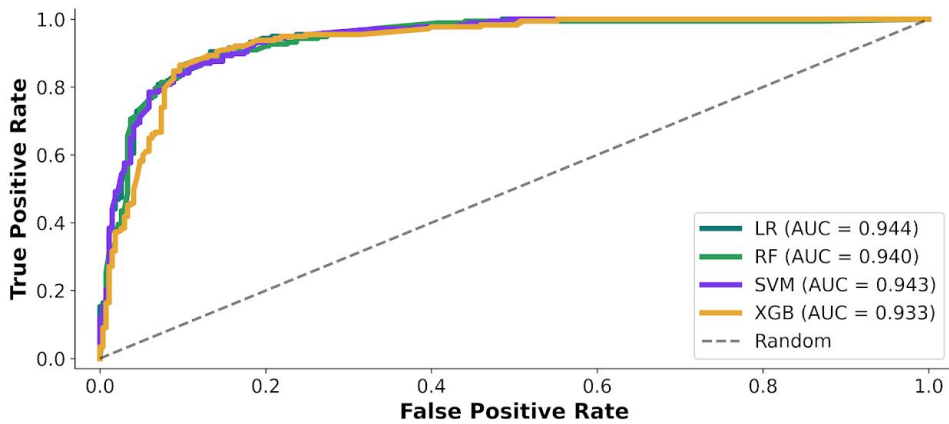
3TC



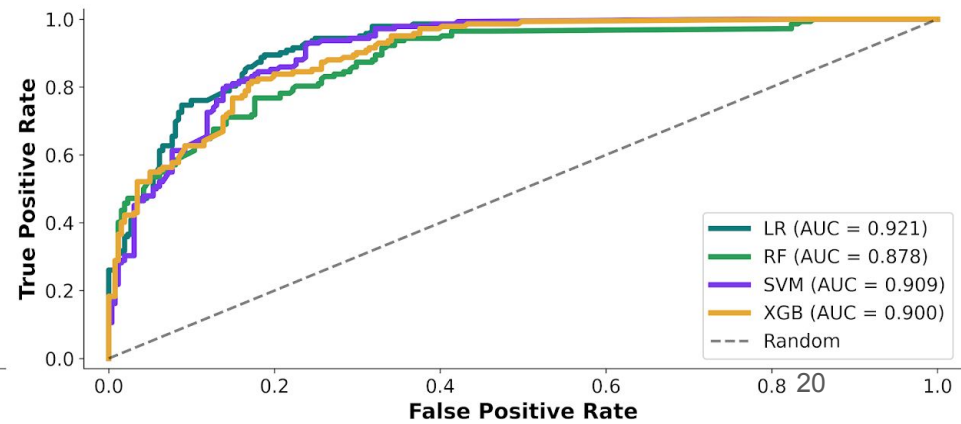
AZT



ABC

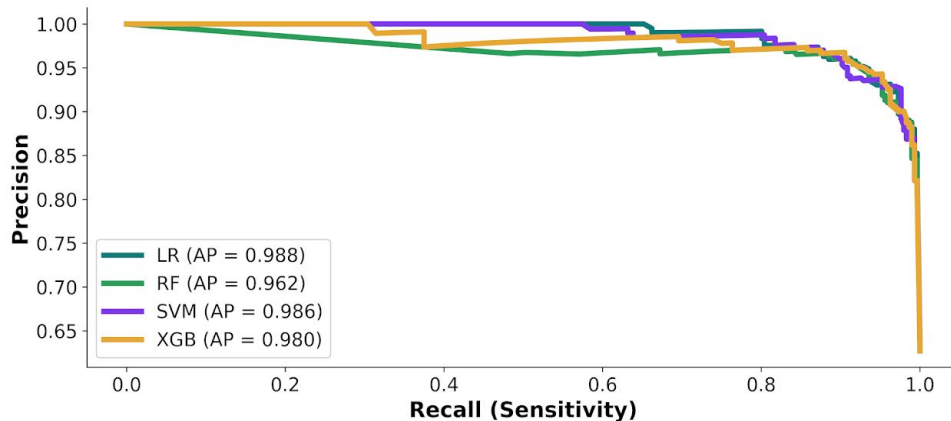


TDF

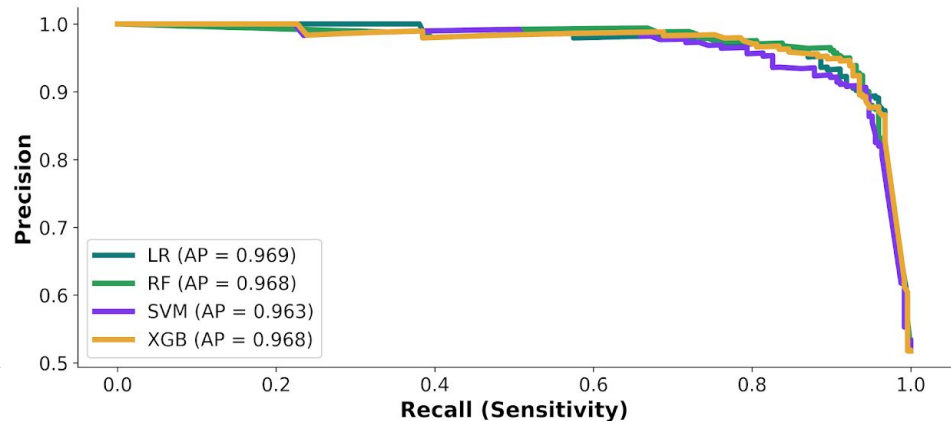


Precision-Recall

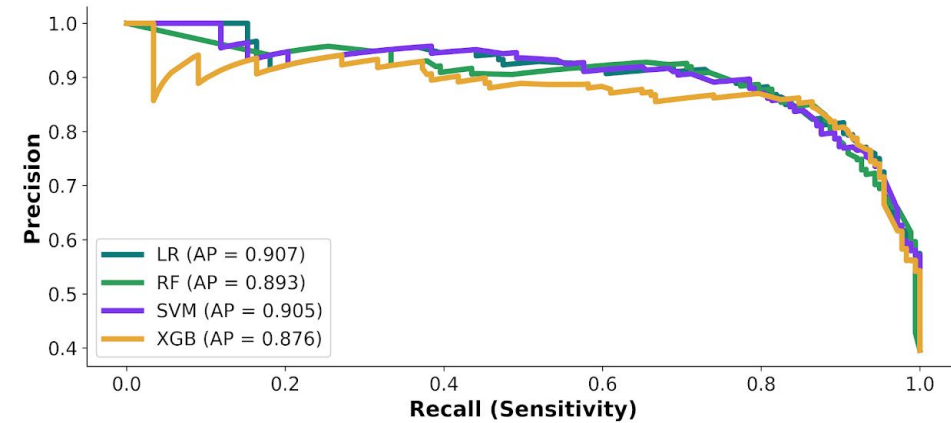
3TC



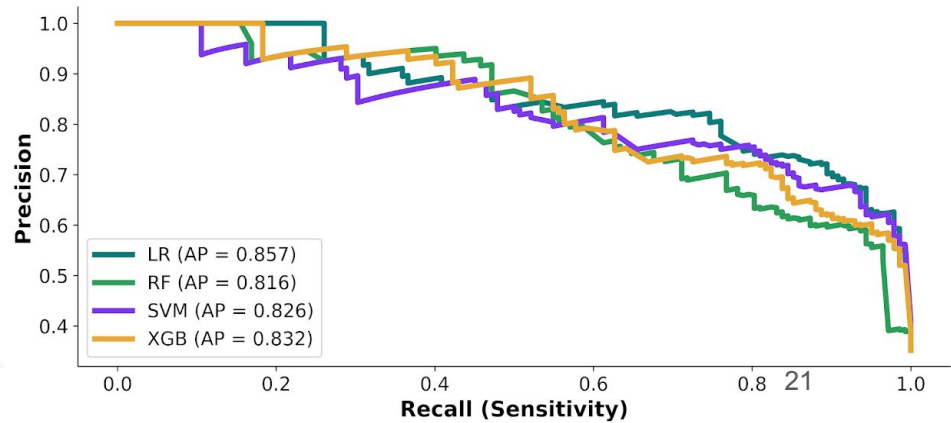
AZT

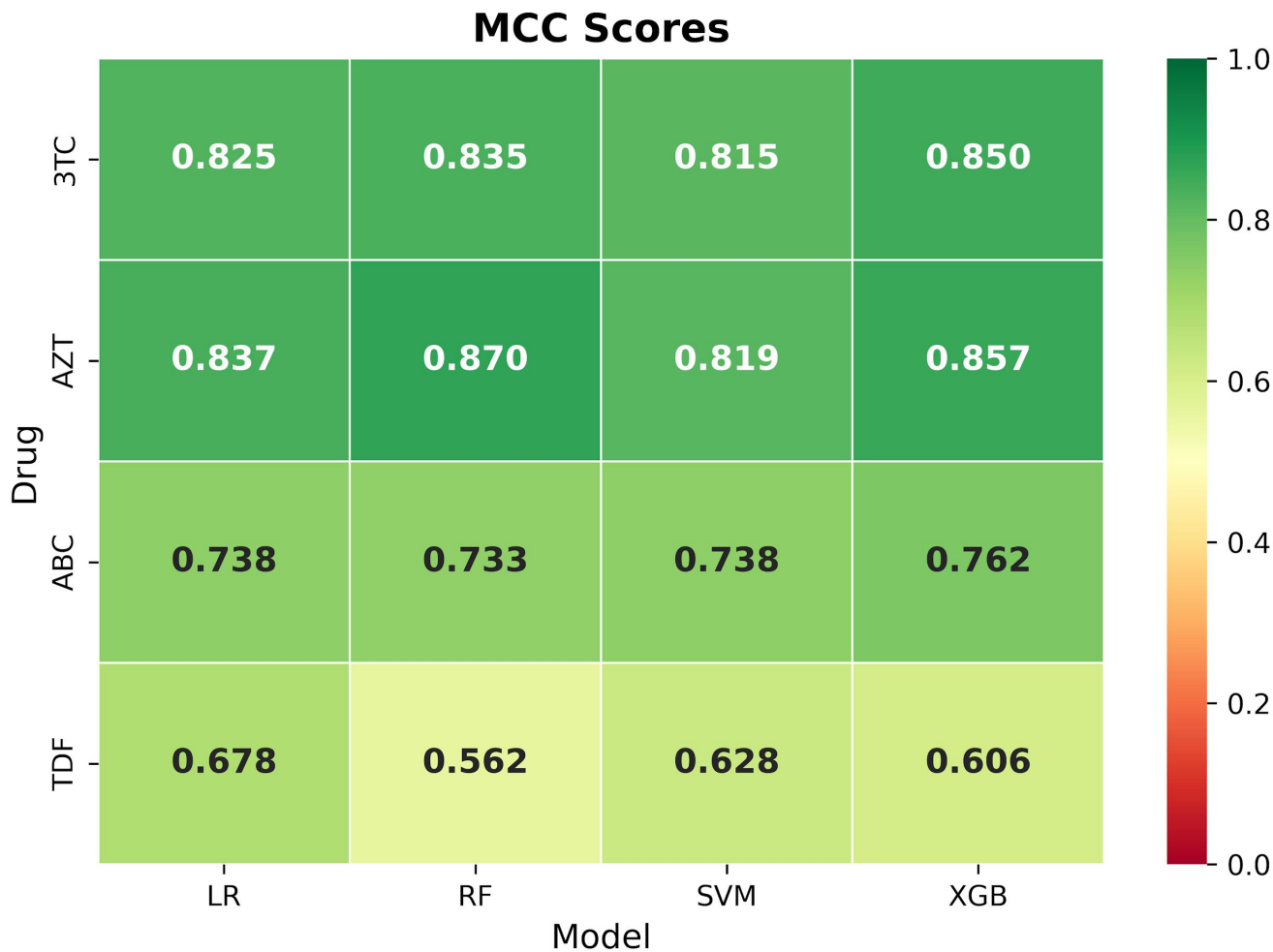


ABC

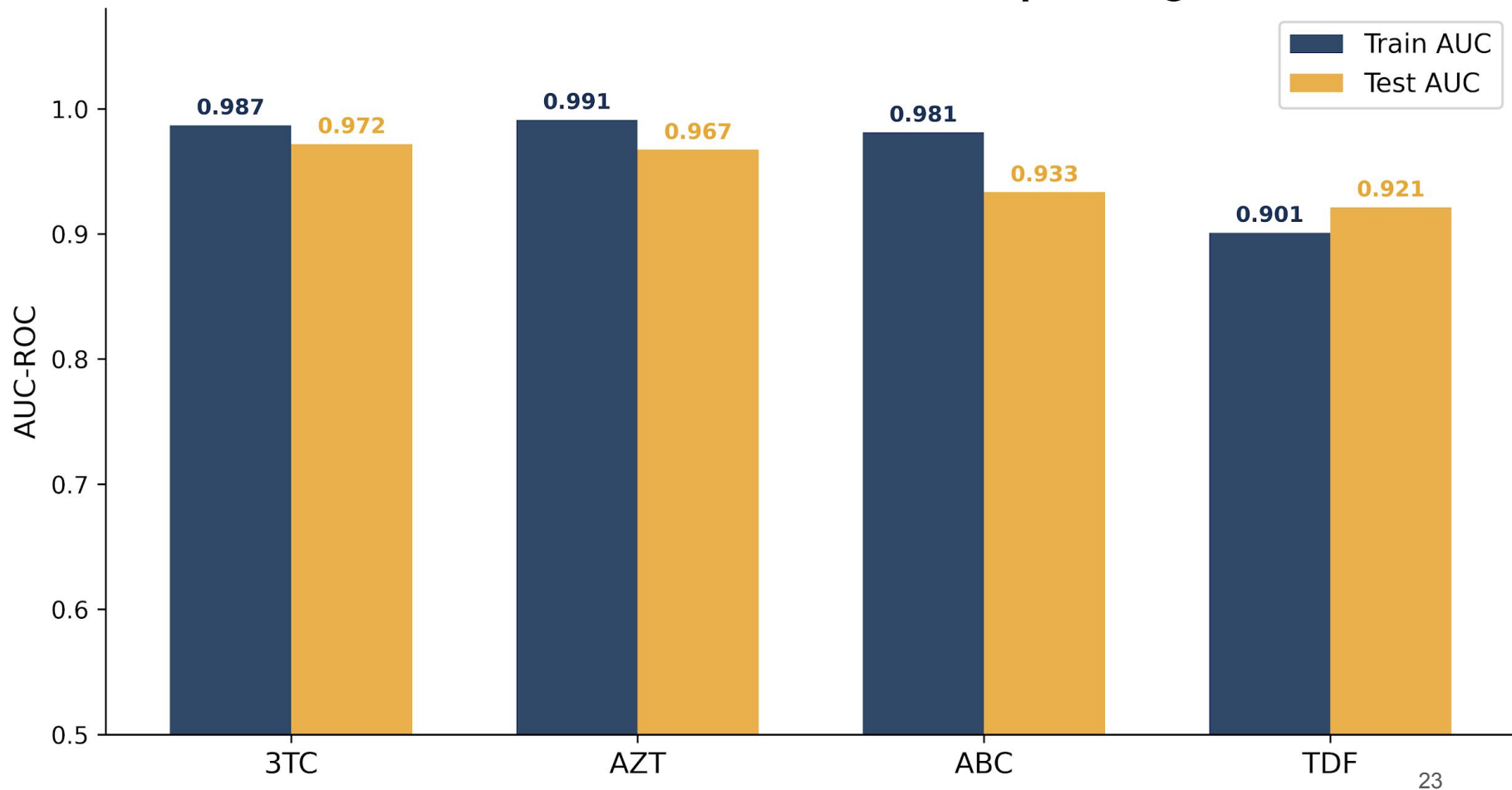


TDF

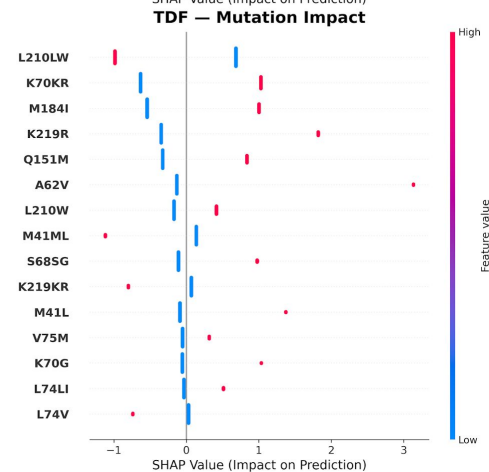
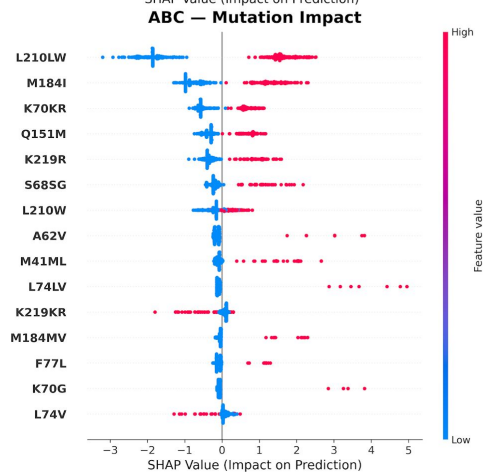
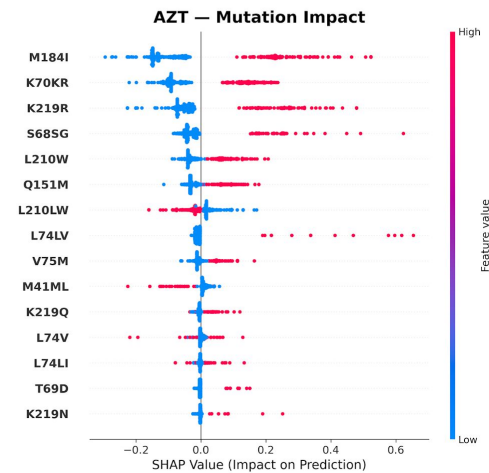
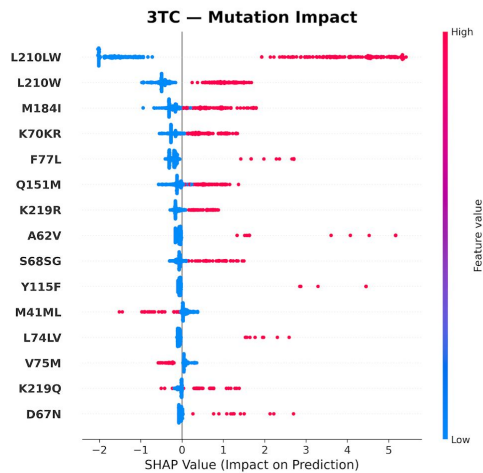




Train vs Test AUC - Best Model per Drug



SHAP Summary Plot





Aim 4: Integration & Deployment



 **DawaFit**

Supported Drugs

Drug	Full Name
3TC	Lamivudine
ABC	Abacavir
AZT	Zidovudine
TDF	Tenofovir

How to use

 **DawaFit**

HIV-1 NRTI Drug Resistance Prediction Tool

Upload an RT FASTA file or paste a nucleotide sequence to predict resistance to first-line NRTI drugs.

 **Input Sequence**

Choose input method

☒ Upload FASTA file ☐ Paste nucleotide sequence

Upload RT FASTA

Developed by | Noella · Getnet · Lucy · Irene | DawaFit · HIV-1 NRTI Drug Resistance Prediction · 2026

[Resimap DawaFit](#)

Aug 6 21:04

0:00

100%

DawaFit

localhost:8501

Deploy

DawaFit

Supported Drugs

Drug	Full Name
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How to use

1. Upload a FASTA file or paste a nucleotide sequence



DawaFit

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Choose input method

☒ Upload FASTA file ☐ Paste nucleotide sequence

Upload RT FASTA

Developed by | Noella · Getnet · Lucy · Irene | DawaFit · HIV-1 NRTI Drug Resistance Prediction · 2026

27



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Asante! Thank You! Erokamano! āmeseginalehu!

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- The ENEZA Residential Data Science Training, class of 2026!

ResiMap Research Group
Mapping Mutations. Predicting Resistance. Enhancing Treatment